



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 251544
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
7/8/2026

Sold To Customer ID TFA Sales Order Purchase Order	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140 34217 331068\1-1 663014514	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
TFA Part ID Description Quantity Customer Part ID	15186 4 XH TEE 1 3709	Heat Code Grade Raw Material Heat Process Mill	A2847 ✓ WPB 4201062DF WPB-CN3R CHANGBAO / ea.

Austenitized @: Held For:	1670° F 60 mins/min	Tempered @: Held For:	° mins/min	Stress Relieved @: Held For:	mins/min	Medium:
------------------------------	------------------------	--------------------------	---------------	---------------------------------	----------	---------

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.18	0.83	0.012	0.005	0.21	0.01	0.04	0.01	0.02	0.00								0.33

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	In-Process Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	52648 (psi)	74404 (psi)	39.80		2"	128	126	129	LONGITUDINAL	STRIP

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

1. THESE FITTINGS WERE COLD FORMED AND NORMALIZED AT 1670F AND AIR COOLED. 2. THESE FITTINGS WERE TENSILE TESTED TO VERIFY PHYSICAL PROPERTIES AFTER THE HEAT TREAT PROCEDURE. 3. THE STARTING MATERIAL IS SEAMLESS PIPE. 4. THE HARDNESS DOES NOT EXCEED 197BHN PART TO PART. 5. THIS PRODUCT IS SEAMLESS. 6. TEES WERE "MAG PARTICLE INSPECTED IN ACCORDANCE WITH ASTM A234 GR WPB.

1) MATERIAL IS MANUFACTURED TO ASTM A234-24 GR WPB, ALSO PER ASME SEC II SA234 GR WPB-2023 EDITION. 2) DESIGN AND DIMENSIONS TO ASME B16.9-2018. 3) MATERIAL MEETS ANSI/NACE MR0175/ISO15156:2015 AND ANSI/ NACE MR0103/ISO17945:2015. 4) MANUFACTURED IN AN ISO 9001 REGISTERED FACILITY, CERTIFICATE #: 30248. 5) TFA, INC. ORGANIZATIONAL STRUCTURE AND CERTIFICATION COMPLIES WITH BS EN 10204-2004 TYPE 3.1.B. 6) TFA INC. FITTINGS SATISFY THE MATERIAL REQUIREMENT OF PRESSURE EQUIPMENT DIRECTIVE (PED) 2014/68/EU, ANNEX 1, PARAGRAPH 4.3. 7) FITTINGS ARE MANUFACTURED IN THE USA. RAW MATERIAL IS FROM DOMESTIC AND GLOBAL SOURCES. 8) FITTINGS WERE MANUFACTURED, SAMPLED, TESTED AND INSPECTED IN ACCORDANCE WITH THE ORDER AND SPECIFICATIONS AND WERE FOUND TO MEET THE REQUIREMENTS.

*UB = Unbroken Test

Heat: A2847 Grade: WPB

Page 1 of 1


Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.